Lecture 32: Introduction to Operational Amplifiers

OPERATIONAL AMPLIFIER

An **operational amplifier** (often **op-amp** or **opamp**) is a DC-coupled high-gain electronic voltage amplifier with a differential input and, usually, a single-ended output. In this configuration, an op-amp produces an output potential (relative to circuit ground) that is typically hundreds of thousands of times larger than the potential difference between its input terminals. Operational amplifiers had their origins in analog computers, where they were used to perform mathematical operations in many linear, non-linear, and frequency-dependent circuits.

* Idea of OP-AMP coined in 1940 to refer special kind of amplifier, that by selection of electronic components, can be configured to perform variety of mathematical operation. It took 27 years to develop a commercial integrated IC of OP-AMP.

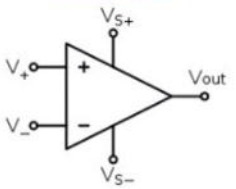
Operational amplifier



A μA741 integrated circuit, one of the most successful operational amplifiers

Type	Discrete circuit Integrated circuit
Invented	Karl D. Swartzel Jr.
First production	1967
Pin configuration	• V+: non-inverting input • V−: inverting input • Vout: output • VS+: positive power supply • VS−: negative power supply The power supply pins (V_{S+} and V_{S-}) can be labeled in different ways (*See IC power supply pins*). Often these pins are left out of the diagram for clarity, and the power configuration is described or assumed from the circuit.

Electronic symbol



Circuit diagram symbol for an op-amp. Pins are labeled as listed above.

The concept underlying the operational amplifier emerged around 1940 in connection with analogue computing. The term operational amplifier was formally introduced in 1947 to describe an amplifier that, with suitable external components and feedback, could perform mathematical operations such as addition, subtraction, integration, and differentiation. The first monolithic IC op-amp, the μA702, was developed in 1963, followed by improved designs such as the μA709 and the widely used μA741. Today, op-amps are available in highly compact IC form, often containing one, two, or four operational amplifiers in a single package.

Year	Development
Around 1940	Op-amp-like amplifier circuits were used at Bell Laboratories for analogue computation.
1947	John R. Ragazzini, Robert H. Randall, and Frederick A. Russell formally introduced the term operational amplifier.
1953	The K2-W vacuum-tube operational amplifier became commercially available.
1963	Bob Widlar developed the μA702 at Fairchild Semiconductor, the first monolithic integrated-circuit op-amp.
1965	The improved μA709 helped establish the practical commercial use of monolithic op-amps.
1968	The μA741 was introduced and became one of the most widely used general-purpose op-amps.
Present day	Op-amps are available as highly compact integrated circuits containing single, dual, or quad amplifiers in one package. Modern op-amps offer features such as very low power consumption, high operating speed, low noise, low offset voltage, rail-to-rail input and output, and operation from low-voltage single

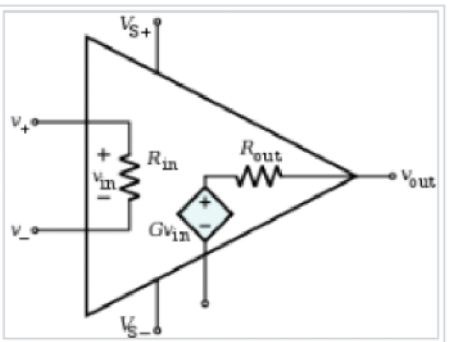
supplies. They are widely used in signal conditioning, sensing, instrumentation, communication, control, audio, biomedical, automotive, and power-electronic systems.

References:

1. **Analog Devices, Op Amp Basics, Op Amp Applications Handbook, Section 1.**
2. **Columbia University Department of Electrical Engineering, The Operational Amplifier.**
3. **Microchip Technology, Operational Amplifier History.**
4. **IEEE Spectrum, Chip Hall of Fame: Fairchild Semiconductor µA741 Op-Amp.**

Characteristics of an Ideal OP AMP

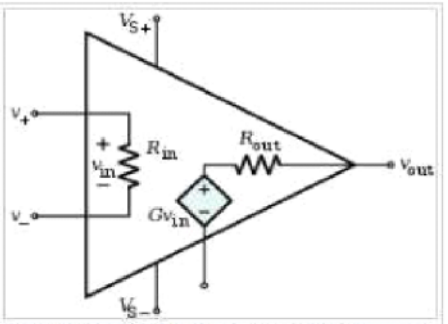
- Infinite open-loop gain G = v_{out} / v_{in}
- Infinite input impedance R_{in}, and so zero input current
- Zero input offset voltage
- Infinite output voltage range
- Infinite bandwidth with zero phase shift and infinite slew rate
- Zero output impedance R_{out}
- Infinite common-mode rejection ratio (CMRR)



An equivalent circuit of an operational amplifier that models some resistive non-ideal parameters.

Characteristics of a Practical OP AMP

- Very large open-loop gain G = v_{out} / v_{in}
- Very large input impedance R_{in}, and so small input current
- Very small input offset voltage
- Large output voltage range
- Finite bandwidth with some phase shift and finite slew rate
- Very small output impedance R_{out}
- Finite common-mode rejection ratio (CMRR)



An equivalent circuit of an operational amplifier that models some resistive non-ideal parameters.

Inside an OP AMP

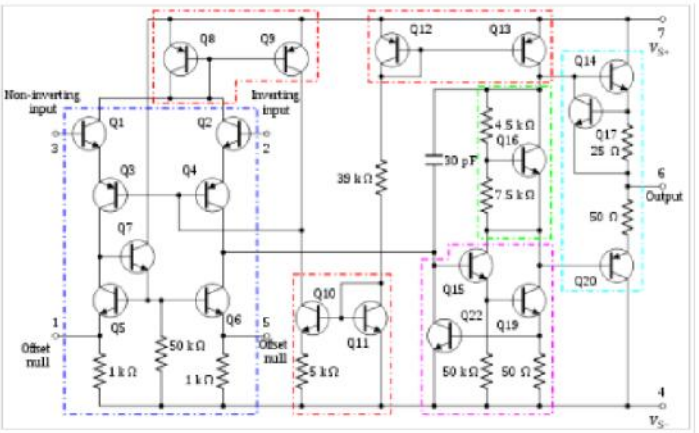
Architecture of OP AMP

A small-scale integrated circuit, the 741 op-amp shares with most op-amps an internal structure consisting of three gain stages.

Differential Amplifier (blue): provides high differential amplification (gain), with rejection of common-mode signal, low noise, high input impedance, and drives a...

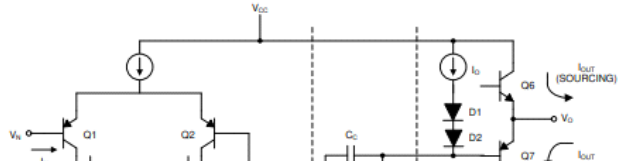
Voltage amplifier (magenta): provides high voltage gain, a single-pole frequency roll-off, and in turn drives the...

Output amplifier (cyan and green) : provides high current gain (low output impedance), along with output current limiting, and output short-circuit protection.

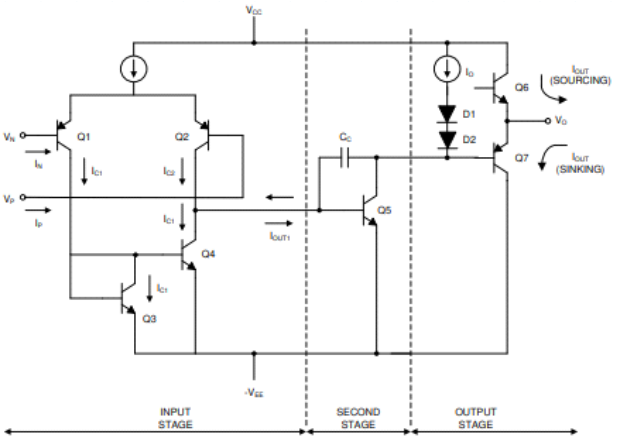


A component-level diagram of the common 741 op-amp. Dotted lines outline: current mirrors, differential amplifier, class A gain stage, voltage level shifter, output stage.

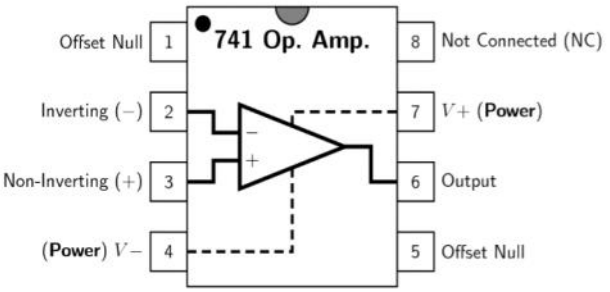
Simplified OP-AMP Circuit Operation



Simplified OP-AMP Circuit Operation

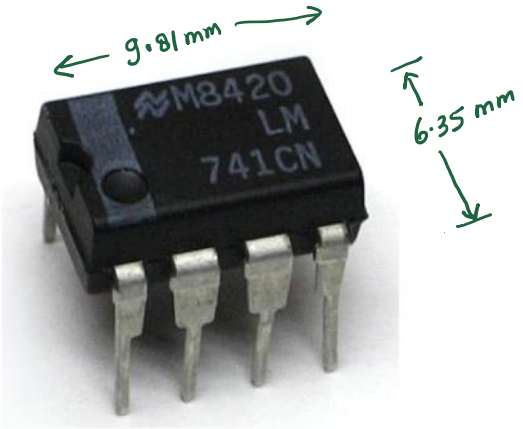


Circuit Configurations



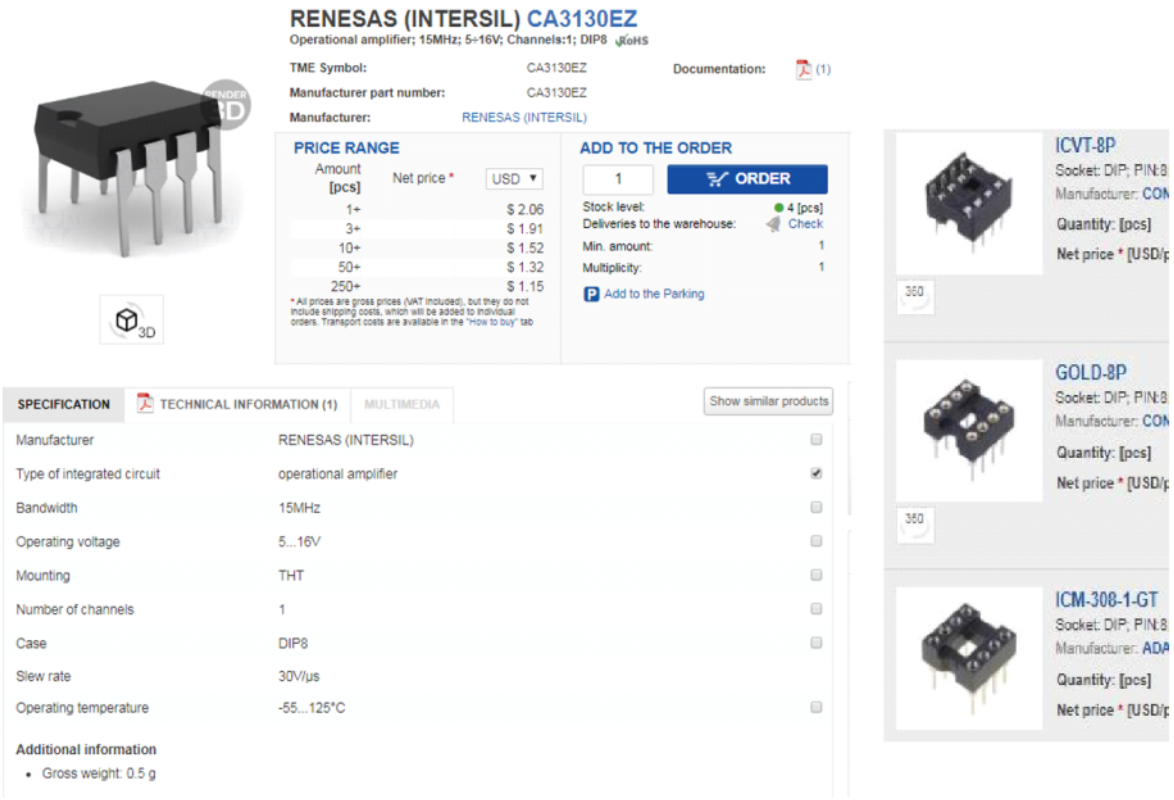


A µA741 integrated circuit, one of the most successful operational amplifiers



9.81 mm × 6.35 mm

IC OP AMP Features



RENESAS (INTERSIL) CA3130EZ

Operational amplifier; 15MHz; 5÷16V; Channels:1; DIP8

TME Symbol: CA3130EZ — Documentation: (1)

Manufacturer part number: CA3130EZ

Manufacturer: RENESAS (INTERSIL)

PRICE RANGE

Amount [pcs]	Net price * USD
1+	$ 2.06
3+	$ 1.91
10+	$ 1.52
50+	$ 1.32
250+	$ 1.15

* All prices are gross prices (VAT included), but they do not include shipping costs, which will be added to individual orders. Transport costs are available in the "How to buy" tab

ADD TO THE ORDER

1 ORDER

Stock level: 4 [pcs]

Deliveries to the warehouse: Check

Min. amount: 1

Multiplicity: 1

Add to the Parking

SPECIFICATION | TECHNICAL INFORMATION (1) | MULTIMEDIA — Show similar products

Manufacturer	RENESAS (INTERSIL)
Type of integrated circuit	operational amplifier
Bandwidth	15MHz
Operating voltage	5...16V
Mounting	THT
Number of channels	1
Case	DIP8
Slew rate	30V/µs
Operating temperature	-55...125°C

Additional information

- Gross weight: 0.5 g

ICVT-8P
Socket: DIP; PIN:8
Manufacturer: CON
Quantity: [pcs]
Net price * [USD/p

GOLD-8P
Socket: DIP; PIN:8
Manufacturer: CON
Quantity: [pcs]
Net price * [USD/p

ICM-308-1-GT
Socket: DIP; PIN:8
Manufacturer: ADA
Quantity: [pcs]
Net price * [USD/p

OP AMP Applications

- Operational amplifiers are the basic building blocks of Analogue electronic circuits.
- They are linear devices with all properties of a DC amplifier.
- We can use external resistors or capacitors to the Op Amp is many different ways to make them different forms of amplifies such as Inverting amplifier, Non inverting amplifier, Voltage follower, Comparator, Differential amplifier, Summing amplifier, Integrator etc. OPAMPs may be single, dual, quad etc. OPAMPs like CA3130, CA3140, TL0 71, LM311 etc have excellent performance with very low input current and voltage.
- The ideal Op Amp has three important terminals in addition to other terminals. The input terminals are Inverting input and Non inverting input.
- The third terminal is the output which can sink and source current and voltage. The output signal is the amplifiers gain multiplied by the value of the input signal.

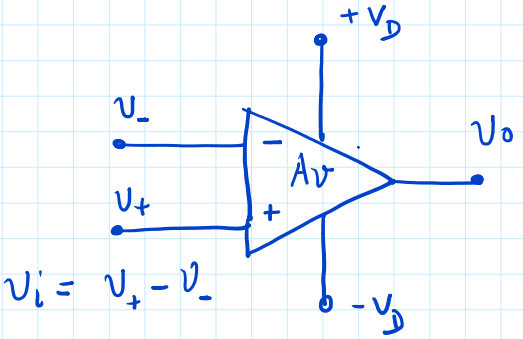
OP AMP Applications

Some more applications...

- audio- and video-frequency pre-amplifiers and buffers
- differential amplifiers
- differentiators and integrators
- filters
- precision rectifiers
- precision peak detectors
- voltage and current regulators
- analog calculators
- analog-to-digital converters
- digital-to-analog converters
- Voltage clamping
- oscillators and waveform generators
- clipper
- clamper(dc inserter or restorer)

OP-AMP : High gain voltage amplifier

$v_o = A_v v_i$ | v_i = Input voltage, v_o = Output voltage, A_v = voltage gain



Features :

	Ideal	Practical
Input resistance, R_i	∞	Very large
Output resistance R_o	0	Very small
Voltage gain	∞	Finite

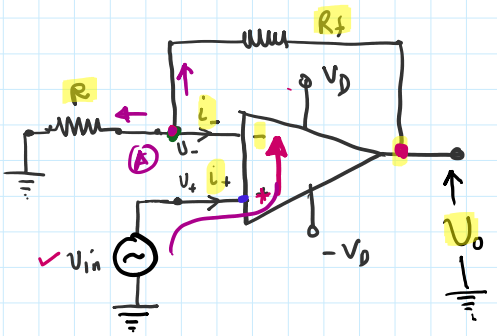
Voltage gain	∞	Finite
Bandwidth	∞	Finite
Voltage balance	Perfect	offsets are presents

Negative Feedback is used: 1) To stabilize the output
2) To have linear gain
3) To reduce sensivity to parameter variations, due environment and ageing.

OP-AMP Operating Modes: 1) Inverting mode
2) Noninverting Mode

Non-Inverting Mode:

$\frac{V_o}{V_{in}} = ? = \text{Voltage gain} = A_v$



Non-Inverting OP-AMP

Concept of virtual ground: Close to ideal amplifier, $i_- \to 0$
$i_+ \to 0$

Open loop gain $V_o = A_v(V_+ - V_-)$

Ideally $A_v \to \infty \Rightarrow$ For very small $(V_+ - V_-)$ is required to produce a finite output voltage.

Hence $(V_+ - V_-) \to 0 \Rightarrow V_+ = V_-$

This can also be understood with concept of Negative Feedback:
i.e. when $V_+ - V_- > 0 \Rightarrow V_+ > V_- \Rightarrow V_o$ increases.
This leads to higher V_-, thus decreasing $(V_+ - V_-)$.
Similarly when $V_+ - V_- < 0, \Rightarrow V_+ < V_-$, leading to reduction in V_o and V_- consequently. This increases $(V_+ - V_-)$. This

Similarly when $v_+ - v_- < 0$, $\Rightarrow v_+ < v_-$, leading to reduction in v_o and v_- consequently. This increases $(v_+ - v_i)$. This in equilibrium $v_+ = v_-$.

Thus we have,

$v_+ = v_{in}$

KCL: $\dfrac{v_- - 0}{R} + \dfrac{v_- - v_o}{R_f}$

$\Rightarrow V_{in}\left(\frac{1}{R} + \frac{1}{R_f}\right) = \frac{v_o}{R_f}$

$\Rightarrow \dfrac{v_o}{v_{in}} = \left(1 + \frac{R_f}{R}\right)$

$$\boxed{v_o = \left(1 + \frac{R_f}{R}\right) v_{in}}$$

$A_v = \left(1 + \frac{R_f}{R}\right)$ if $R_f = 250\ k\Omega$, $R = 10\ k\Omega$

$A_v = \left(1 + \frac{250}{10}\right) = 26$

Inverting Mode Operation of OP-AMP

$\dfrac{v_o}{v_{in}} = ?$

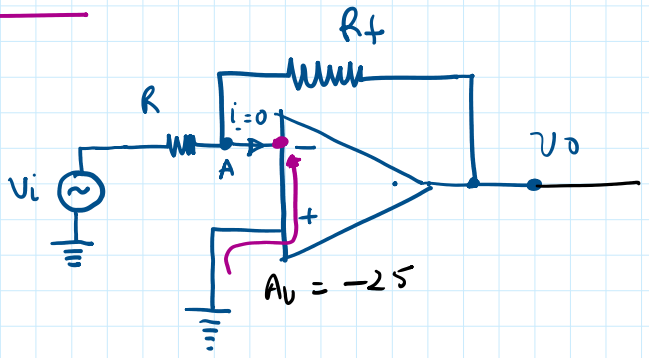
$\dfrac{0 - v_i}{R} + \dfrac{0 - v_o}{R_f} = 0$

$$\Rightarrow \boxed{\frac{v_o}{v_i} = A_v = -\frac{R_f}{R}}$$

$R_f = 250\ k\Omega$

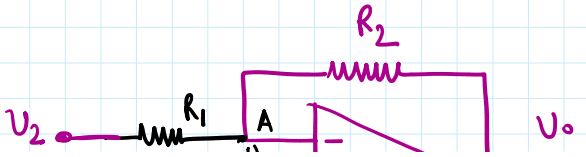
$R = 10\ k\Omega$

$$\boxed{A_v = -25}$$

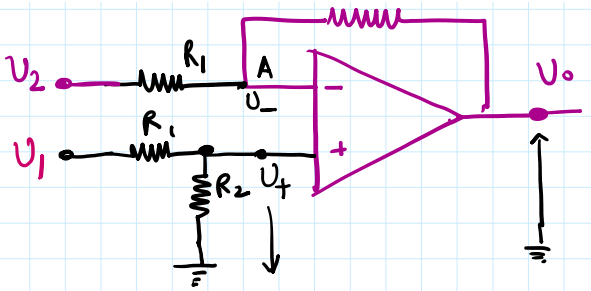


Difference Amplifier:

$v_o = f(v_1, v_2)$



$$V_0 = f(V_1, V_2)$$



$$V_+ = \left(V_1 \times \frac{R_2}{R_1+R_2}\right)$$

$$\frac{V_- - V_2}{R_1} + \frac{V_- - V_0}{R_2} = 0$$

$$V_1 * \frac{R_2}{(R_1+R_2)} \left(\frac{R_1+R_2}{R_1 R_2}\right) = \frac{V_2}{R_1} + \frac{V_0}{R_2}$$

$$\frac{V_0}{R_2} = -\frac{V_2}{R_1} + \frac{V_1}{R_1} = \frac{1}{R_1}(V_1 - V_2)$$

$$\boxed{V_0 = \left(\frac{R_2}{R_1}\right)(V_1 - V_2)}$$